

Designing Spiking Neural Network on Embedded Systems with On-Chip Learning Ability

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Abstract

In this study, I propose an object recognition method using Spiking Neural Networks (SNNs), which mimic biological neural activity and are expected to be highly energy-efficient as a next-generation AI technology. Specifically, the objective is to classify with high accuracy whether an object is a "hand" or "not a hand" using images captured by a thermal camera as input. For the training of the SNN, we employed Backpropagation Through Time (BPTT), which can handle temporal information. As a preprocessing step, low-resolution thermal images of 24×32 pixels were converted into time-series spike trains using rate coding, which generates spikes according to pixel intensity. Furthermore, during training, the loss was calculated based on the membrane potential values of the output layer at each time step, allowing the network to directly learn time varying information. Experimental results demonstrated that the proposed method achieved a classification accuracy of 90.0% for hand detection on both a GPU and a Raspberry Pi 5. Furthermore, the average inference time on the Raspberry Pi was approximately 29.5 ms, indicating that the system is capable of real-time processing on embedded devices.

1 Introduction

The human brain is known to perform highly complex and sophisticated cognitive processing while consuming only about 20 watts of power [1, 7, 9]. Spiking Neural Networks (SNNs), which aim to emulate this highly efficient biological information-processing mechanism, have attracted considerable attention as a next-generation low-power artificial intelligence technology. SNNs transmit information using pulse-like spikes and represent information through the temporal patterns of these spikes, resulting in computational principles that differ fundamentally from those of conventional neural networks. [2, 7]

In this study, I employ Backpropagation Through Time (BPTT) as a learning method for SNNs, enabling the handling of temporal information. As an evaluation task, I address a binary classification problem that determines whether a hand is present in thermal camera images. The input data consist of low-resolution thermal images with a resolution of 24×32 pixels, which are converted into time-series spike trains using rate cod-

ing. During training, the loss is computed based on the membrane potentials of the output neurons at each time step, allowing the model to directly learn temporal features.

2 Method

2.1 Overview of the System

This section presents an overview of the thermal image classification system based on a spiking neural network (SNN) proposed in this study. The system takes images captured by a thermal camera as input and aims to determine whether a human hand is present in the image.

First, thermal images with a resolution of 24×32 pixels are acquired. These low-resolution images contain temperature information and can be processed efficiently even in environments with limited computational resources. The captured images are normalized and then converted into time-series spike trains, allowing static image data to be handled by the SNN. The system is implemented using the `snnTorch` library [8].

Next, the generated spike trains are input to the SNN and processed over multiple time steps. The SNN consists of an input layer, a hidden layer, and an output layer, and it extracts features from the thermal images by integrating spike information over time.

Finally, a classification result is produced based on the activity of the neurons in the output layer, determining whether a hand is present or absent in the input image.

2.2 Dataset and Preprocessing

In this study, I utilized a dataset consisting of low-resolution thermal images captured by infrared sensors. The resolution of each image is 24×32 pixels, classified into two categories: "Hand" and "Not Hand".

2.3 Input Encoding

To allow the SNN to handle static thermal images, this study represents each pixel value as the number of spikes generated over time. In this approach, larger values are expressed by more frequent spikes, while smaller values produce fewer spikes.

Each thermal image is first normalized to a range between 0 and 1. The normalized pixel values are then treated as the probability of generating a spike at each time step. Pixels with higher values generate spikes

more frequently, whereas pixels with lower values generate spikes less often. The simulation is performed over 25 time steps.

Because information is represented by the overall number of spikes rather than exact timing[4], this method provides a stable representation even when randomness is present. In addition, the conversion from image values to spike generation is simple and easy to implement, making it suitable for learning in the proposed SNN model.

2.4 Spiking Neuron Model

In spiking neural networks (SNNs), information processing is performed by spiking neurons, which are simplified models of biological neurons. A spiking neuron integrates incoming signals over time and emits a spike when certain conditions are satisfied.

In this study, the Leaky Integrate-and-Fire (LIF) model is used as the spiking neuron model.[3] The LIF model is widely adopted in SNN research due to its simple structure and low computational cost.

At discrete time step t , the membrane potential V_t is updated based on the previous membrane potential and the input spike as follows:

$$V_t = \beta V_{t-1} + Z_t \quad (1)$$

Here, Z_t represents the input spike (or input current) at time t , and β ($0 < \beta < 1$) is a decay factor that models the leakage of the membrane potential. This equation allows the neuron to integrate inputs while gradually losing its potential over time.

When the membrane potential exceeds a predefined threshold V_{th} , the neuron generates an output spike. The spiking behavior is described by the following equation:

$$S_t = \begin{cases} 1 & (V_t \geq V_{th}) \\ 0 & (V_t < V_{th}) \end{cases} \quad (2)$$

In this equation, S_t denotes the output spike at time t . After a spike is generated, the membrane potential is reset as follows:

$$V_t \leftarrow 0 \quad (3)$$

Through this mechanism, the LIF neuron integrates input spikes over time and produces an output spike only when sufficient input has accumulated. In this study, LIF neurons are used to process spike sequences converted from thermal images, enabling the model to capture temporal characteristics in the data.

2.5 Network Architecture

In this study, I employ a spiking neural network (SNN) that takes low-resolution thermal images of 24×32 pixels

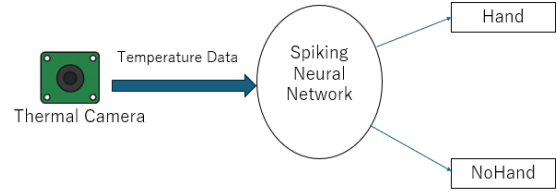


Figure 1: The overall architecture of the proposed hand detection system.

(768 pixels in total) acquired by a thermal camera as input. The proposed model has a three-layer architecture consisting of an input layer, a hidden layer, and an output layer. The input layer comprises 768 input neurons, each corresponding to a pixel in the thermal image. The hidden layer consists of 256 Leaky Integrate-and-Fire (LIF) neurons, which receive input spikes through a fully connected layer and integrate membrane potentials over time to generate spikes at each time step. The output layer is composed of two LIF neurons corresponding to the “Not Hand” and “Hand” classes, respectively, and the final classification is performed based on the membrane potentials of the output neurons at each time step.

2.6 Learning Method

In this study, I use Backpropagation Through Time (BPTT) as a learning method to handle temporal information[5]. BPTT is a training approach designed for time-dependent data, in which the model updates its parameters by evaluating the outputs along the time axis[6].

During training, the input spike sequences are processed over multiple time steps, and the output states at each time step are used to compute the error. The final loss is obtained by averaging the errors over all time steps, which allows the model to learn from information distributed across time rather than relying on a single moment.

2.7 Data Collection

In this study, I used a low-resolution thermal camera with a resolution of 24×32 pixels. To construct the dataset, I captured thermal images under various conditions. The data acquisition included cases where a hand was held in front of the camera (“Hand”) and cases where no hand was present (“Not Hand”). To ensure the robustness of the model, the “Not Hand” category includes not only empty spaces but also images containing other hot or cold objects.

The collected dataset was divided into three subsets:

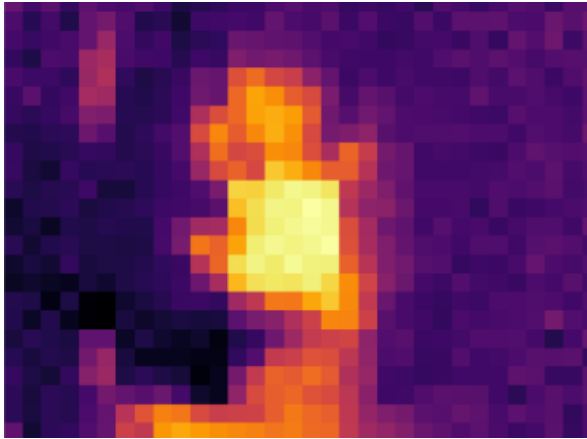


Figure 2: Thermal image of a hand.

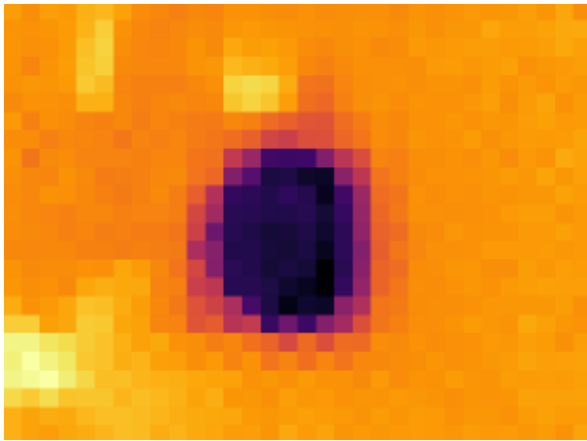


Figure 3: Thermal images of cold objects other than hands.

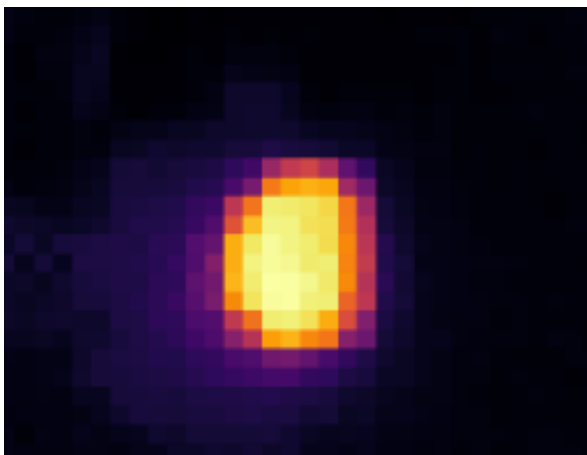


Figure 4: Thermal images of hot objects other than hands.

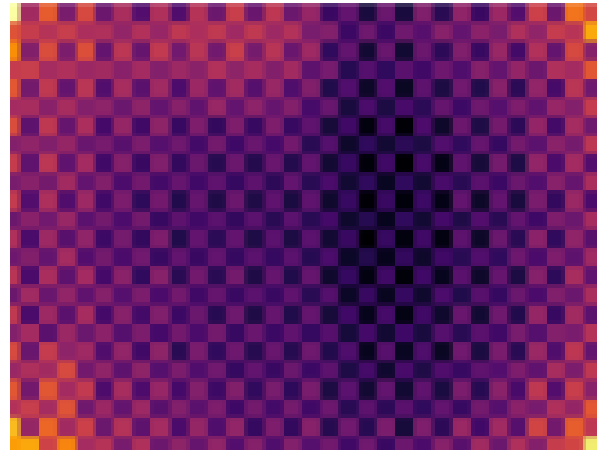


Figure 5: Example thermal image of an empty space classified as "not a hand."

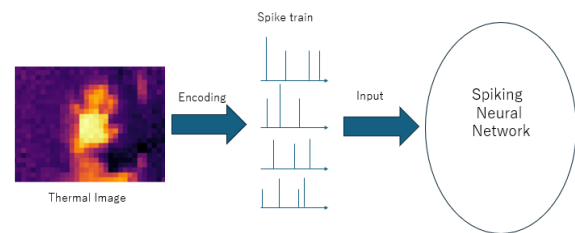


Figure 6: The encoding process converting thermal image pixel intensity into spike trains. Brighter pixels (higher temperature) generate spikes with higher frequency.

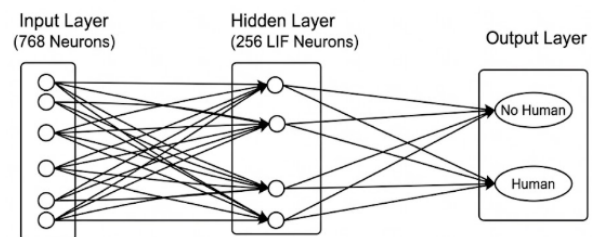


Figure 7: The architecture of the Spiking Neural Network used in this study. It consists of an input layer (768 neurons), a fully connected hidden layer (256 LIF neurons), and an output layer (2 neurons) for binary classification.

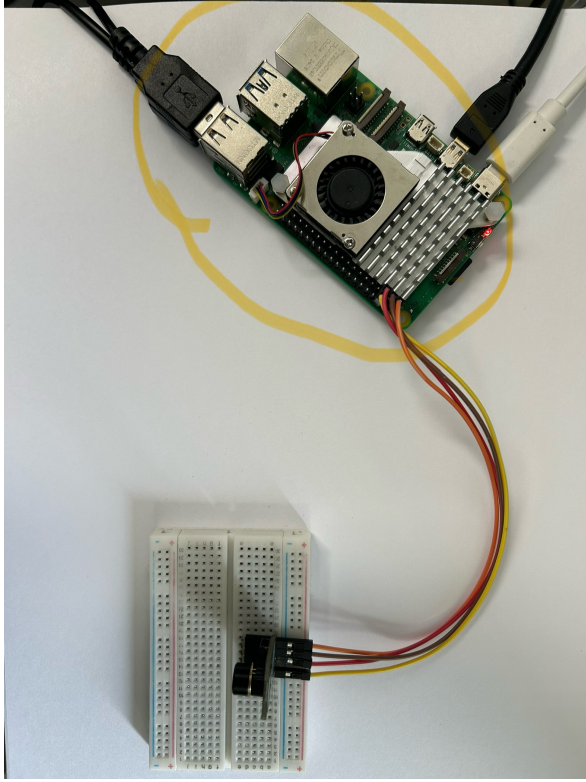


Figure 8: The experimental setup for the embedded system. The Raspberry Pi 5 (circled in yellow) is connected to the MLX90640 thermal camera via the I2C interface.

training, validation, and test sets. I collected 699 images for training, 150 images for validation, and 299 images for testing to evaluate the model's performance. Table 1 summarizes the number of images used in this study.

Table 1: Number of images in the dataset.

Dataset Type	Number of Images
Training	699
Validation	150
Test	299

3 Result

3.1 Classification Accuracy

Tables 2 and 3 show the classification accuracy for Hand and Not Hand samples on GPU and Raspberry Pi environments.

Table 2: Classification accuracy on GPU

Ground Truth	Hand (%)	Not Hand (%)
Hand	90.0% (269)	10.0% (30)
Not Hand	27.1% (81)	72.9% (218)

Table 3: Classification accuracy on Raspberry Pi

Ground Truth	Hand (%)	Not Hand (%)
Hand	90.0% (269)	10.0% (30)
Not Hand	25.8% (77)	74.2% (222)

3.2 Inference Speed

Table 4 compares the average inference time between GPU and Raspberry Pi environments.

Table 4: Average inference time comparison

Environment	Avg. Inference Time (ms)
GPU	4.73
Raspberry Pi	29.51

3.3 Confidence for Hand Samples

The confidence values for Hand samples were analyzed separately to evaluate the reliability of the classification results.

Table 5: Average confidence for Hand samples on GPU

Metric	Value
Number of samples	299
Average confidence	0.9404

Table 6: Average confidence for Hand samples on Raspberry Pi

Metric	Value
Number of samples	299
Average confidence	0.9367

Comparison between devices: When comparing the results between the GPU and the Raspberry Pi, the average confidence values were almost identical (0.9404 vs 0.9367). This indicates that the proposed SNN model maintains its reliability even on an embedded device with limited computational resources, despite the significant difference in hardware specifications.

3.4 Confidence for Not Hand Samples

Confidence values for Not Hand samples were also evaluated to analyze misclassification tendencies.

Comparison between devices: A similar trend was observed for the NoHuman samples. The confidence

Table 7: Average confidence for Not Hand samples on GPU

Metric	Value
Number of samples	299
Average confidence	0.8904

Table 8: Average confidence for Not Hand samples on Raspberry Pi

Metric	Value
Number of samples	299
Average confidence	0.8687

scores on the Raspberry Pi (0.8687) were comparable to those on the GPU (0.8904). This suggests that the misclassification tendencies are due to the model's characteristics (e.g., reacting to thermal patterns) rather than hardware limitations or calculation errors on the device.

4 Discussion

In this study, a Spiking Neural Network (SNN) was constructed for binary classification of thermal images into Hand and Not Hand classes using supervised learning with Backpropagation Through Time (BPTT). The proposed method was evaluated in both a GPU environment and a Raspberry Pi environment, and classification accuracy as well as inference speed were compared.

4.1 Comparison of Classification Accuracy between GPU and Raspberry Pi

The experimental results show that, in both the GPU and Raspberry Pi environments, the classification accuracy for images containing hands (Hand) reached approximately 90%. In particular, the accuracy for the Hand class was almost identical in both environments. This result indicates that comparable classification performance can be achieved even on an embedded platform such as the Raspberry Pi, similar to that obtained in the GPU environment.

On the other hand, for images without humans (Not Hand), the proportion of samples misclassified as Hand was relatively high, at approximately 25–27%. This trend was consistently observed in both environments, suggesting that the misclassification is not caused by differences in hardware performance, but rather by characteristics of the model and the input data.

4.2 Analysis of False Positives Caused by Hot Objects

An analysis of misclassified Not Hand samples revealed that false detections were mainly caused by high-temperature objects that were not hands. In thermal

images, the presence of a hand is primarily represented as a region with relatively high temperature. Therefore, the model is considered to have learned temperature intensity and localized high-temperature patterns as important features.

As a result, when Not Hand objects with high temperatures were present in the input, they were sometimes incorrectly classified as Hand. This finding suggests that the proposed model does not yet distinguish humans based on semantic shape information, but rather detects thermal patterns that resemble those of humans.

This observation represents an important insight of this study, indicating that human detection using thermal images is inherently challenging when relying solely on temperature information.

4.3 Inference Speed and Applicability to Embedded Systems

In terms of inference speed, the average inference time was approximately 4.8 ms in the GPU environment and approximately 29.5 ms in the Raspberry Pi environment. Although inference on the Raspberry Pi was about six times slower than on the GPU, an inference time of around 30 ms per frame is still practical for many real-time embedded applications.

Furthermore, since the classification accuracy was largely preserved on the Raspberry Pi, these results demonstrate that SNN-based thermal image classification can be effectively deployed on resource-constrained embedded systems.

5 Conclusion

This study proposed a Spiking Neural Network for hand detection using thermal images and applied supervised learning based on Backpropagation Through Time. The performance of the proposed method was evaluated in both GPU and Raspberry Pi environments to investigate its applicability to embedded systems.

The experimental results demonstrated that high classification accuracy of approximately 90% was achieved for images containing hands, with no significant performance difference between the GPU and Raspberry Pi environments. In contrast, false detections were observed for images without hands, particularly in cases involving high-temperature non-human objects, highlighting a key challenge specific to thermal-image-based human detection.

These results indicate that relying solely on temperature information is insufficient to completely eliminate false positives, and that additional information or processing strategies are required for more robust detection.

As future work, methods to reduce false detections caused by hot objects should be investigated. Such im-

provements are expected to contribute to the realization of a more reliable thermal-based human detection system suitable for deployment in embedded environments.

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